Group and solo flights

Technical report on the FFVL cross-country archive

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Question. Do paragliders move differently when they fly in a group? If they do, is it the group, or who flies in groups, where and when they fly, or the moment they choose to fly together?
Answer. Over a quarter of an hour the progress in a group is straighter: $\Delta H_{\mathrm{loc}}(1000) = +0.055$ raw, $+0.037$ within the same flight, $+0.043$ at the same site-day and hour, and only $+0.003$ from who flies in groups. It is not the zone (reweighting removes at most 19% of it) and not the moment (on the same flight it starts with the group, $+0.027$, and stops with it, $+0.031$). Within one glide or climb (100 s) nothing changes.
Also in a group, once stratified: more time climbing, more searches that end in a climb, a higher climb rate, faster and flatter glides, a higher cross-country speed.
Inside the group. Most of the effect is there with two pilots; at the same place and hour only 7+ stands apart. The pilot ahead flies a little faster but finds no more thermals.
Limits. Strata and windows fix pilot, day, site, hour, zone and the drift of the flight, not what brings pilots together; a contact is shared air within sight; phases exist on 1 Hz flights only. Data: 38.1-min contacts of the note *Group flights in free flight*, 154 598 paraglider flights (21.7% in a group at some point).

1 What we compare

Which flights, which populations, which statistics? The groups are those of the note *Group flights in free flight*; this section fixes what the report measures on them.

- **Flights.** 156 391 paraglider and 6 094 hang-glider flights on 6 209 days, cleaned tracks of the old repository (soaring, tag `legacy-final`). Each flight is analysed after its *departure* (first fix 3 km from the takeoff); before it, the pilot is in the *local phase* and counts as neither solo nor group. Analyses use loggers of at most 10 s; phases (climb, search, transition=glide) are Vilpellet's runs and exist on 1 Hz flights only (70% of the paragliders).
- **Grouping.** The method, its null model and its parameters are in the note *Group flights in free flight* (`ideas/group-identification`): definitions in its sections 3–4, calibration and tuning in sections 5–6. This report uses its run with $\delta = 2$ min, $\varepsilon_h = 500$ m, $\varepsilon_z = 300$ m, $\theta_{\max} = 45°$, $g = 2$ min and $q = 0.1$, hence $\tau^* = 38.1$ min. A contact is *shared air within sight*, not proof of interaction.
- **Populations. S** solo flights; **G** group time, the stretches in a configuration (a fixed set of flights in one group over a maximal interval); **A** the *same flights alone*, the stretches of group flights outside any configuration: the control with the same pilots, gliders, loggers and days. G_n is the group time in configurations of n pilots: **"n pilots" always means the size of the configuration at that instant.**
- **TA-MSD.** Positions on a 10 s grid of the flight clock; per flight $\overline{\delta^2}(\tau) = \sum \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2 / \#\text{origins}$ pooled over its stretches, ensemble MSD = mean over flights (equal weight); stretches of at least 1250 s, 21 lags from 10 to 1000 s. H: half the slope of $\log$ MSD against $\log \tau$ over all lags ($H = 1$ straight line). $H_{\mathrm{loc}}(\tau)$: the same slope on the lags within ± 0.25 decades of τ (Appendix A).
- **Phase statistics.** Ratios of sums over runs (time in climb, searches that end in a climb, climb rate, glide speed, glide ratio; cross-country speed over 30 min). Five of twelve are drawn: the others are explained by them (Appendix C).
- **Strata.** *Zone*: cell of 25 km of the middle of a piece of track; *site-day*: flights of one day with takeoffs within 3 km; *hour*: local hour.
- **Error bars** are 5–95% bootstrap bands resampling whole dependency *blocks*, the flights tied by sharing episodes (never single flights), and recomputing the statistic: blocks of the note for pooled statistics (1000 draws); blocks joining episodes and site-days for stratified ones (200 draws, same draws for all). They

measure block sampling only, not the choice of q or the tolerances, and two bands that do not overlap are not a test; differences are computed draw by draw.

What the grouping gives. Of the 154 598 paraglider flights with a logger of at most 10 s, 33 501 (21.7%) fly in a group at some point; of the 6 039 hang-glider flights, 418 (6.9%). Group time is 12.8% of all paraglider flight time after the departure, and on average 45% of the time of a group flight. The 36 575 pair intervals last 40 to 96 min (10th to 90th percentile; median 52 min) and form 40 051 configurations in 17 169 episodes. Half of the episodes have two flights, 1% have more than 9, the largest has 53. By the size of the configuration, the paraglider group time is: 2 pilots 64%, 3 pilots 19%, 4 pilots 8%, 5-6 pilots 6%, 7+ pilots 3% (the largest configuration has 27 pilots). In contact, one pilot is behind the other by a median of 20 s (75% within 45 s).

2 The raw difference

How do group time and solo flights differ before any control?

	solo flights S	group time G	same flights alone A
flights (TA-MSD)	119 129	33 435	29 611
H (10–1000 s)	0.899 [0.898, 0.899]	0.916 [0.915, 0.917]	0.914 [0.913, 0.915]
$H_{\mathrm{loc}}(100\ \mathrm{s})$	0.919 [0.919, 0.919]	0.927 [0.927, 0.928]	0.932 [0.931, 0.932]
$H_{\mathrm{loc}}(1000\ \mathrm{s})$	0.837 [0.837, 0.838]	0.893 [0.891, 0.894]	0.860 [0.859, 0.862]
time in climb (%)	30.8 [30.7, 30.9]	33.0 [32.8, 33.2]	29.1 [28.9, 29.2]
searches ending in a climb (%)	43.3 [43.2, 43.4]	49.5 [49.2, 49.7]	45.9 [45.7, 46.1]
climb rate (m/s)	1.22 [1.22, 1.22]	1.26 [1.25, 1.27]	1.33 [1.32, 1.34]
glide speed (km/h)	30.1 [30.1, 30.2]	33.3 [33.1, 33.5]	32.8 [32.6, 32.9]
glide ratio	10.4 [10.4, 10.4]	12.9 [12.8, 13.0]	11.3 [11.2, 11.4]
cross-country speed (km/h)	17.2 [17.2, 17.3]	20.0 [19.8, 20.3]	20.0 [19.8, 20.1]

Table 1: Pooled statistics of S, G and A (paragliders), with 5–95% bands (blocks of the note). TA-MSD on loggers of at most 10 s, phases on 1 Hz flights.

Group time progresses straighter over long lags than solo flights, $\Delta H_{\mathrm{loc}}(1000) = +0.055$; it also spends more time climbing, ends more searches in a climb and glides faster (Table 1; the curves over every lag are in Appendix D). The same flights alone sit between the two at long lags and have the highest climb rate, so part of the raw difference may belong to who flies in groups, and to where and when.

What we keep. A difference to explain: how much of it is the group?

3 Group, pilots, place or moment?

Besides the group, four things differ between group time and solo flights: the pilots who fly in groups, the day and the site, the hour, and the moment pilots choose to fly together. Three tests take them away:

- five contrasts inside strata separate the pilots (selection) from the group, at the same site-day and hour and within the same flight (Section 3.1);
- reweighting the solo flights to the zones, months and hours of the groups tests the place (Section 3.2);
- the same flight before, during and after a group tests the moment (Section 3.3).

3.1 Pilots or group: five contrasts

Question. How much of the difference belongs to who flies in groups, and how much to the group? **Method.** Five contrasts, each computed inside strata and averaged over those with both populations (weights $n_P n_Q/(n_P + n_Q)$, n the flights of each side): G − S pooled; G − S and A − S (*selection*) at the same site-day and hour; G − A within the flight and at the same site-day and hour (the *group effect*). The last two fail in opposite ways: the first compares a flight with itself at another time of day, and group time falls around midday while the time alone of the same flights falls earlier and later (section 7 of the note); the second compares different pilots at the same time. Where the two agree, the effect belongs to the group.

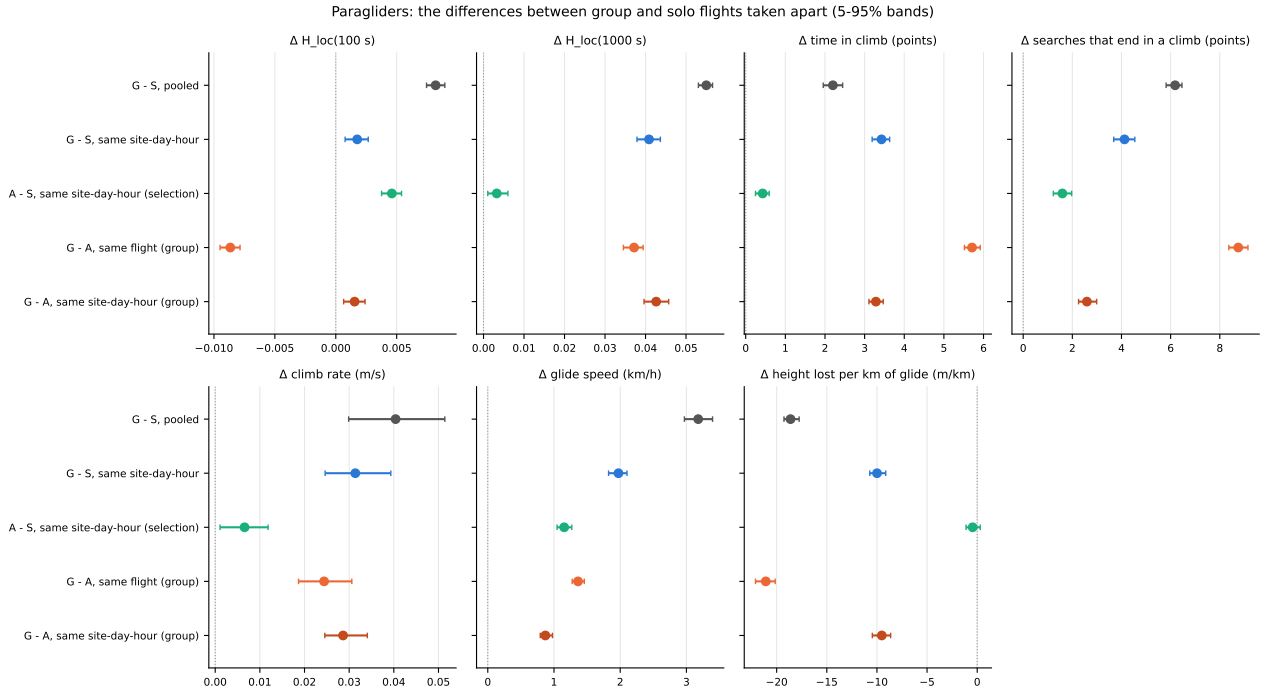


Figure 1: Each statistic (panel) for the five contrasts (rows), with 5–95% bands. Shares of time in points; lower height lost per km is a better glide.

Δ	G − S pooled	G − S same s-d-h	A − S same s-d-h	G − A same flight	G − A same s-d-h
$H_{\text{loc}}(100\text{ s})$	+0.008 [+0.007, +0.009]	+0.002 [+0.001, +0.003]	+0.005 [+0.004, +0.005]	−0.009 [−0.009, −0.008]	+0.002 [+0.001, +0.002]
$H_{\text{loc}}(1000\text{ s})$	+0.055 [+0.053, +0.057]	+0.041 [+0.038, +0.044]	+0.003 [+0.001, +0.006]	+0.037 [+0.035, +0.039]	+0.043 [+0.040, +0.046]
time in climb (%)	+2.20 [+1.96, +2.45]	+3.43 [+3.19, +3.63]	+0.43 [+0.25, +0.59]	+5.71 [+5.52, +5.92]	+3.28 [+3.11, +3.47]
searches to climb (%)	+6.18 [+5.82, +6.46]	+4.12 [+3.68, +4.54]	+1.60 [+1.23, +1.97]	+8.75 [+8.37, +9.15]	+2.59 [+2.25, +2.99]
climb rate (m/s)	+0.040 [+0.030, +0.051]	+0.031 [+0.025, +0.039]	+0.007 [+0.001, +0.012]	+0.024 [+0.019, +0.031]	+0.029 [+0.025, +0.034]
glide speed (km/h)	+3.18 [+2.97, +3.39]	+1.97 [+1.82, +2.10]	+1.15 [+1.05, +1.27]	+1.36 [+1.27, +1.46]	+0.87 [+0.79, +0.98]
height lost per km (m/km)	−18.6 [−19.3, −17.8]	−10.0 [−10.7, −9.1]	−0.5 [−1.1, +0.3]	−21.1 [−22.1, −20.1]	−9.5 [−10.5, −8.6]
cross-country speed (km/h)	+2.81 [+2.59, +3.01]	+2.25 [+2.09, +2.39]	+1.31 [+1.21, +1.43]	+0.70 [+0.60, +0.78]	+1.29 [+1.16, +1.41]

Table 2: The contrasts of Figure 1: difference and 5–95% band (s-d-h: same site-day and hour).

Reading.

- **Straighter progress over 1000 s** survives every control: $\Delta H_{\text{loc}}(1000)$ is +0.055 pooled, +0.037 within the flight, +0.043 at the same site-day and hour; selection gives only +0.003.
- **Nothing within one phase (100 s):** −0.009 within the flight, +0.002 at the same place and hour; the pooled +0.008 was selection and composition.
- **Climbing:** more time climbing (+5.71, +3.28 points), more searches that end in a climb (+8.75, +2.59 points), a higher climb rate (+0.024, +0.029 m/s). Pooled, the climb rate of G is below that of A: the pooled ratios weight flights by their time in each population.
- **Gliding:** faster and flatter glides, +1.36, +0.87 km/h and −21.1, −9.5 m/km less height lost per km; selection adds +1.15 km/h of glide speed (group pilots glide faster even alone). Cross-country speed: +0.70, +1.29 km/h.

What we keep. With the pilot, the day, the site and the hour fixed, a group progresses straighter over a quarter of an hour, climbs more and glides faster and flatter; who flies in groups explains little of it.

3.2 The place: zones, months and hours

Question. Do groups fly where, and when, any pilot would progress in a straight line? **Method.** Each piece of track is assigned to a *stratum*: zone (cell of 25 km of its middle), zone×month, zone×hour, site-day×hour.

In every stratum with both populations the ensemble MSD of S (and of A) is *reweighted to the mix of G over the strata*: S then flies, in the statistic, where and when G flies. We compare $H_{\mathrm{loc}}(\tau)$ over every lag and the fitted H; "as is" is the same comparison without the reweighting, on the same strata (a flight counts once per cell it crosses).

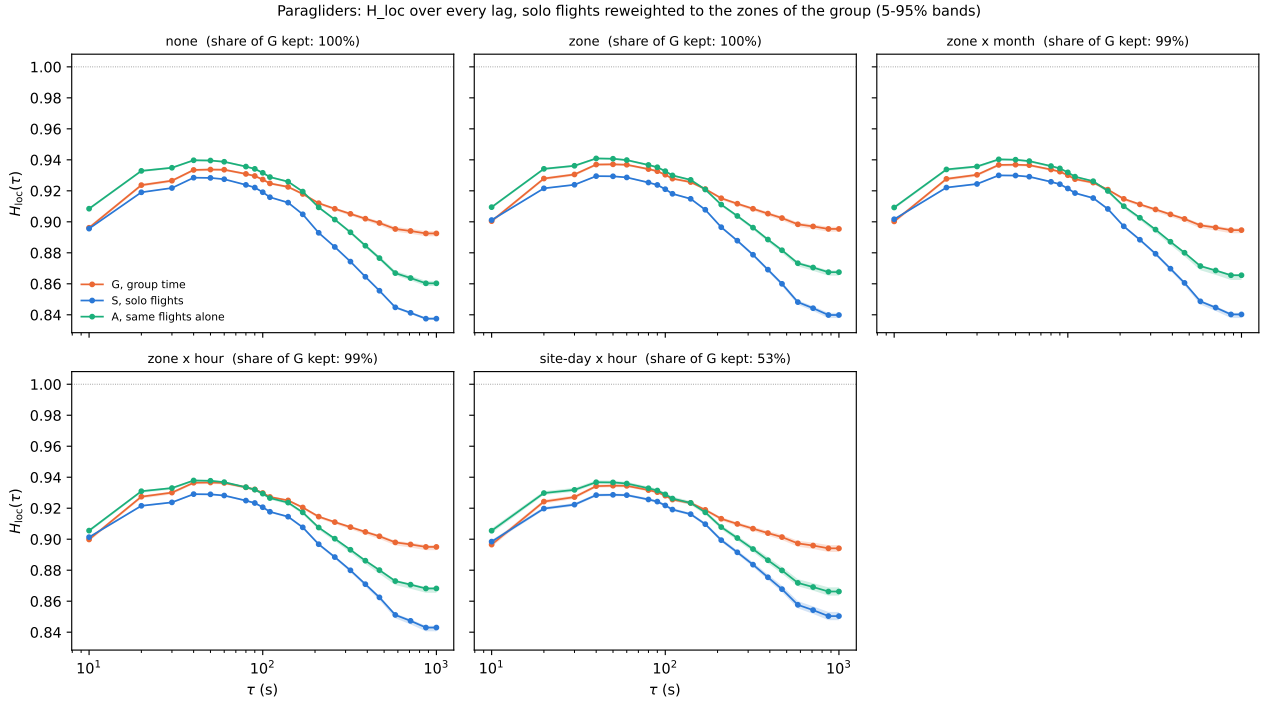


Figure 2: $H_{\mathrm{loc}}(\tau)$ of G (orange), of S reweighted to the strata of G (blue) and of A reweighted to them (green), 5–95% bands. First panel: all flights pooled (the raw comparison). "Share of G kept": flights of G with a solo counterpart in their stratum.

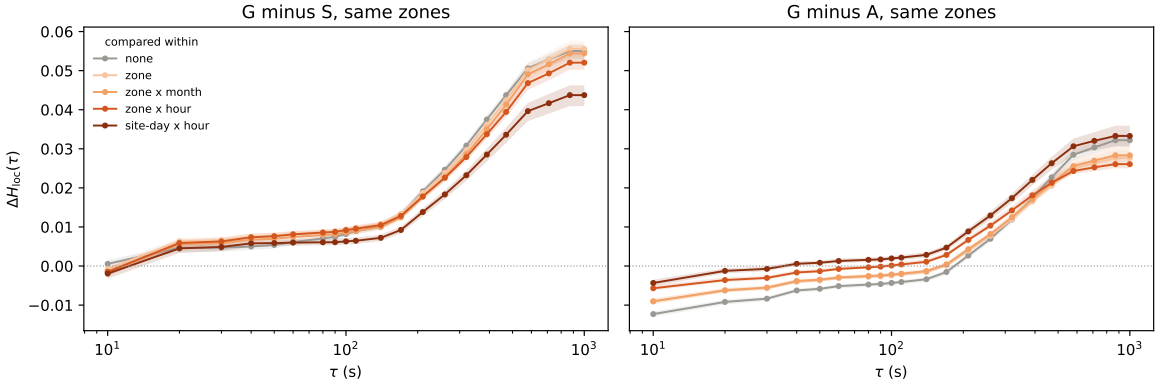


Figure 3: $\Delta H_{\mathrm{loc}}(\tau)$ over every lag: G minus S (left) and G minus A (right), the other population reweighted to the strata of G; one line per stratification, finer strata in darker orange.

		fitted H (10–1000 s)			ΔH			$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$		
compared within	G kept (%)	G	S | G	A | G	G − S as is	G − S reweighted	G − A reweighted	G − S as is	G − S reweighted	G − A reweighted
none	100	0.916	0.899	0.914	+0.017	+0.017 [+0.016, +0.018]	+0.002 [+0.001, +0.003]	+0.055	+0.055 [+0.053, +0.057]	+0.032 [+0.030, +0.034]
zone	100	0.919	0.902	0.916	+0.021	+0.018 [+0.017, +0.019]	+0.003 [+0.003, +0.004]	+0.059	+0.056 [+0.054, +0.058]	+0.028 [+0.026, +0.029]
zone × month	99	0.919	0.902	0.915	+0.022	+0.017 [+0.016, +0.018]	+0.003 [+0.003, +0.004]	+0.064	+0.054 [+0.053, +0.057]	+0.028 [+0.027, +0.031]
zone × hour	99	0.919	0.902	0.913	+0.022	+0.017 [+0.016, +0.018]	+0.005 [+0.005, +0.006]	+0.064	+0.052 [+0.050, +0.054]	+0.026 [+0.025, +0.028]
site-day × hour	53	0.917	0.904	0.913	+0.015	+0.013 [+0.012, +0.014]	+0.008 [+0.007, +0.009]	+0.051	+0.044 [+0.041, +0.046]	+0.033 [+0.031, +0.036]

Table 3: Fitted H and differences by stratification. "Reweighted": the other population reweighted to the strata of G (value, 5–95% band); "as is": the same strata without reweighting. $H_{\mathrm{S|G}}$: H of S reweighted to G.

Reading.

- **The gap opens above $\tau \approx 150$ s.** ΔH_{loc} between G and S is +0.005 to +0.011 from 20 to 140 s and $+0.055$ at 1000 s, above one climb–glide cycle: the group acts on how cycles are chained, not on a single glide or climb.
- **The zone removes little.** Reweighting S to the cells of G lowers $\Delta H_{\mathrm{loc}}(1000)$ by 6% ($+0.059 \rightarrow +0.056$) and the fitted ΔH by 15%; with the month, 15% and 23%; with the hour, 19% and 24%. Solo pilots fly where groups fly: 100% of G has a solo counterpart in its cell.
- **Same site-day and hour** (53% of G kept): $\Delta H_{\mathrm{loc}}(1000) = +0.044$ [$+0.041$, $+0.046$], 14% below its as-is value; fitted $\Delta H = +0.013$ [$+0.012$, $+0.014$]. The fitted H of S rises by a few thousandths when reweighted (0.899 $\rightarrow$ 0.904) and never reaches that of G (0.917).
- **Against the same flights alone:** $\Delta H_{\mathrm{loc}}(1000)$ is $+0.032$ pooled and $+0.028$, $+0.026$, $+0.033$ in zone, zone$\times$hour and site-day$\times$hour strata. Below 150 s the sign of G $-$ A depends on the hour, which is why the two estimates of Section 3.1 are read together.

Limits. Cells are 25 km, from the middle of a piece; the terrain inside a cell is not controlled, and a group may choose better air within it: this is part of what we call the group effect. Paragliders only.

What we keep. The place and the hour where groups fly explain at most a fifth of the gap; the rest stays at the same site, day and hour.

3.3 The moment: the same flight before, during and after

Question. Pilots may fly together in the moments and places where the progress is straight anyway. If so, the same flight would already be straight just before the group forms and still just after it breaks. **Method.** Every stretch of group time lasting at least 50 min is an *event* with up to four windows of 50 min of the flight clock: *before* (in the solo time of the same flight), *start* and *end* of the group, *after*. Before and after need a solo stretch of that length next to the group. For two windows the ensemble MSD is taken on the events that have both. **Controls:** two consecutive windows with no change of state, in the solo time of the group flights (A) and in solo flights (S): the drift of a flight between consecutive windows.

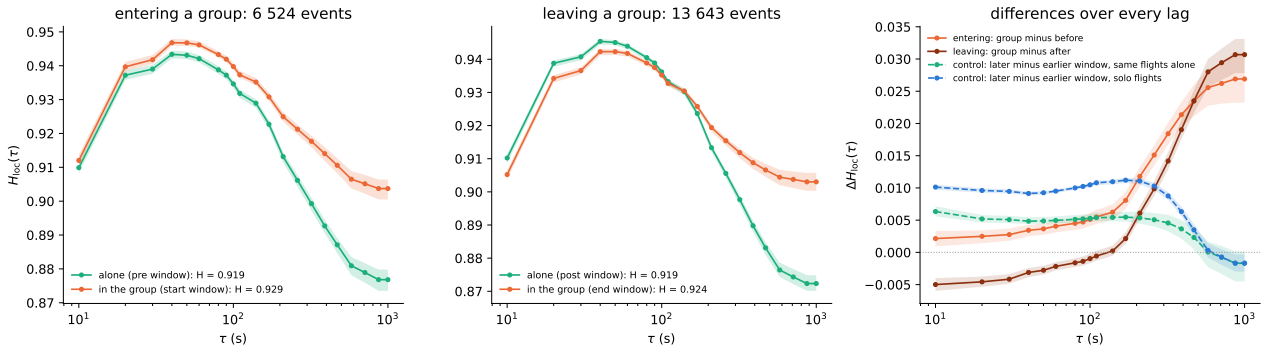


Figure 4: $H_{\mathrm{loc}}(\tau)$ of the same flights alone (green) and in the group (orange), entering (left) and leaving (middle), 5–95% bands. Right: the differences, with the two controls (later minus earlier window; dashed).

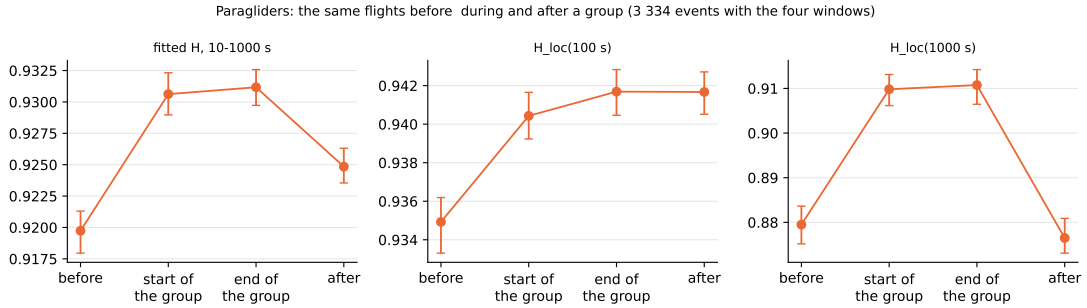


Figure 5: The four windows on the events that have all of them. Start and end overlap when the group lasts less than 2×50 min.

comparison	events	ΔH	$\Delta H_{\mathrm{loc}}(100)$	$\Delta H_{\mathrm{loc}}(1000)$
entering: start − before	6 524	+0.010 [+0.009, +0.011]	+0.005 [+0.004, +0.006]	+0.027 [+0.023, +0.031]
leaving: end − after	13 643	+0.005 [+0.004, +0.006]	−0.001 [−0.002, −0.000]	+0.031 [+0.028, +0.033]
after − before	6 320	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.007]	−0.003 [−0.007, +0.001]
control A: later − earlier	16 998	+0.005 [+0.004, +0.005]	+0.005 [+0.005, +0.006]	−0.002 [−0.005, +0.000]
control S: later − earlier	69 648	+0.009 [+0.008, +0.009]	+0.010 [+0.010, +0.011]	−0.002 [−0.003, −0.000]

Table 4: Differences of the fitted H and of H_{loc} at 100 and 1000 s (value, 5–95% band) between windows. Controls: two consecutive windows with no change of state, in the solo time of the group flights (A) and in solo flights (S). Events: those with both windows.

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	4 786	+0.7 [+0.3, +1.1]	+0.0 [−1.1, +1.0]	+0.046 [+0.034, +0.062]	+0.92 [+0.69, +1.15]	−2.3 [−3.5, −1.3]
leaving: end − after	10 097	+1.5 [+1.2, +1.9]	+1.2 [+0.4, +1.9]	−0.007 [−0.017, +0.003]	+0.52 [+0.35, +0.64]	−10.9 [−11.8, −10.0]
after − before	4 718	−1.4 [−1.8, −1.0]	−2.8 [−4.0, −1.6]	+0.043 [+0.024, +0.059]	+0.33 [+0.12, +0.55]	+9.1 [+7.8, +10.4]
control A: later − earlier	12 568	−1.1 [−1.3, −0.9]	−1.4 [−2.0, −0.8]	+0.018 [+0.009, +0.027]	+0.07 [−0.05, +0.20]	+3.3 [+2.5, +4.2]
control S: later − earlier	49 092	−1.6 [−1.7, −1.5]	−2.3 [−2.7, −2.0]	+0.011 [+0.008, +0.015]	+0.04 [−0.03, +0.10]	+4.9 [+4.5, +5.3]

Table 5: The phase statistics of the same comparisons (differences with 5–95% band); runs and glides are placed in the window that holds their middle.

Reading.

- **Straightness follows the group.** $\Delta H_{\mathrm{loc}}(1000)$ is +0.027 [+0.023, +0.031] entering and +0.031 [+0.028, +0.033] leaving, against −0.002 and −0.002 in the controls.
- **Before and after are alike:** after minus before is −0.003 [−0.007, +0.001]. On the 3 334 events with all four windows $H_{\mathrm{loc}}(1000)$ is 0.879 before, 0.910 and 0.911 in the group, 0.876 after; the step matches the within-flight estimate of Section 3.1 (+0.037).
- **Nothing within one glide or climb:** at 100 s +0.005 entering and −0.001 leaving, inside the drift of the controls (+0.005, +0.010). The fitted H (+0.010, +0.005 against +0.005, +0.009) does not separate from the drift, because its slope is dominated by the lags below 150 s.
- **Glides are faster and flatter in the group, both ways:** glide speed +0.92 km/h entering and +0.52 leaving, height lost per km −2.3 and −10.9 m/km, against +0.04 and +0.07 km/h, +3.3 and +4.9 m/km in the controls. Time in climb, successful searches and climb rate move by the size of the drift: no robust change.

Limits. The windows fix the pilot, the glider, the day and the slow drift of the flight, not the moment: what brings pilots together (a convergence line, a ridge, a common route) may last as long as the group and straighten the progress by itself, and two pilots must move alike to stay within 500 m for 50 min. Events are long groups with solo flight beside them (6 524 entries, 13 643 exits); phases on 1 Hz flights only. Windows of 21 min: Appendix G.

What we keep. On the same flight the straighter progress starts with the group and stops with it.

4 Inside the group: size and leader

Does the effect grow with the number of pilots, and does the pilot ahead fly differently from the one behind?

4.1 Size of the configuration

Method. The group time is cut where the configuration changes, so every stretch has a fixed number of pilots n (2, 3, 4, 5–6, 7+); a flight can contribute to several sizes. Two contrasts by size: each size minus the same flights alone (within the flight), and each size minus configurations of 2 pilots at the same site-day and hour.

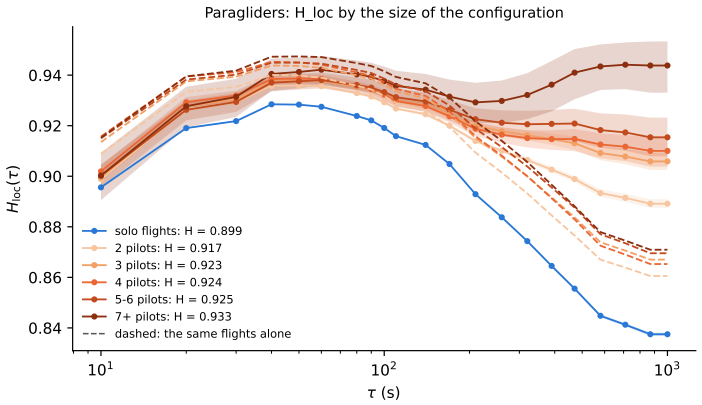


Figure 6: $H_{\text{loc}}(\tau)$ over every lag by the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

		H		$H_{\text{loc}}(1000\text{ s})$						
pilots	flights	G_n	alone	G_n	alone	time in climb (%)	searches to climb (%)	climb rate (m/s)	glide speed (km/h)	glide ratio
solo	119 129	0.899		0.837		30.8	43.3	1.22	30.1	10.4
2	26 411	0.917	0.914	0.889	0.861	32.7	49.0	1.27	33.2	12.5
3	9 480	0.923	0.919	0.906	0.867	33.6	49.9	1.26	33.5	13.1
4	4 073	0.924	0.920	0.910	0.865	33.6	51.0	1.25	33.5	13.7
5–6	2 686	0.925	0.922	0.915	0.870	33.1	50.1	1.23	33.1	14.1
7+	1 142	0.933	0.924	0.944	0.871	34.6	51.7	1.20	34.6	15.9

Table 6: Paragliders, pooled by the size of the configuration ("alone": the same flights alone).

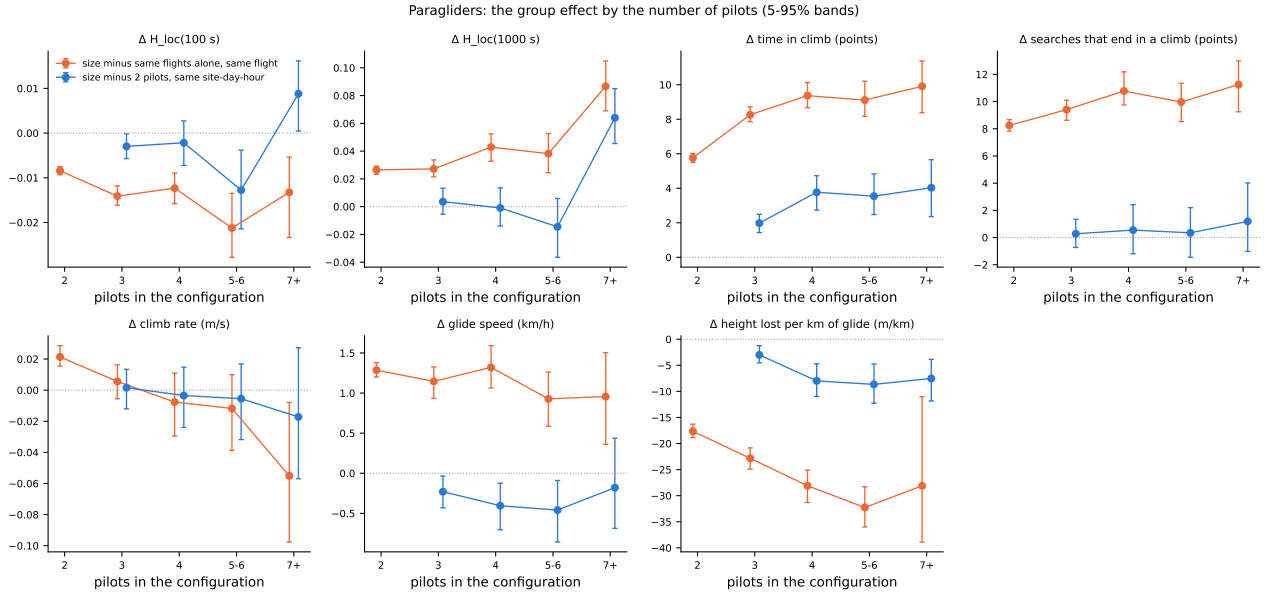


Figure 7: Orange: $G_n - A$ within the flight. Blue: $G_n - G_2$ at the same site-day and hour.

Reading. Pooled, H and H_{loc} rise with n while the same flights alone stay flat (Fig. 6, Table 6), and within the flight the effect grows with n (+0.026 with 2 pilots, +0.087 with 7+). At the same site-day and hour, configurations of 3 to 6 pilots do not differ from pairs ($\Delta H_{\text{loc}}(1000)$ between −0.015 and +0.004, bands across zero): the rise of the pooled curves belongs to the days and sites where large groups form. Only 7+ stands apart (+0.064 [+0.045, +0.085], cross-country speed +7.0 km/h), a rare configuration (3% of group time). Time in climb and the flatter glide do grow with n at the same place and hour (+2.0 to +4.0 points; −3 to −9 m/km).

4.2 Leader and follower

Method. In configurations of two pilots, the signed median delay ℓ_{med} of the pair interval (Appendix B of the note) says who passes first: the leader, when the other passes at least 15 s later; otherwise the two fly abreast

(35% of the pairs; median delay 20 s). Leader and follower are compared within the same configuration (Appendix H). How groups form and break and whether the same pilots lead are in section 7 of the note.

> **Reading.** The pilot ahead flies a little faster (glide speed $+0.86$ [$+0.74$, $+0.99$] km/h, climb rate $+0.021$ [$+0.014$, $+0.028$] m/s) but does not find thermals more often (searches ending in a climb -0.01 [-1.03, $+0.83$] points, height lost per km $+1.6$ [-0.9, $+4.5$] m/km): the follower shows no sign of using the leader to find the next thermal.

What we keep. Most of the effect is there with two pilots; beyond a pair, size adds climbing and flatter glides but not straighter progress, except in the rare configurations of 7 or more. Leading changes speed, not the finding of thermals.

5 Conclusions and limits

Supported

- the progress over a quarter of an hour is straighter in a group, with the pilot, the day, the site and the hour fixed;
- no effect within one glide or climb (100 s) at the same hour;
- on the same flight the straighter progress over 1000 s starts with the group and stops with it (before $\approx$ after);
- more time climbing, more searches that end in a climb, faster and flatter glides;
- climb rate and cross-country speed are *higher* in a group once the composition is controlled (the pooled readings reversed);
- selection of pilots is small for the motion;
- the pilot ahead flies a little faster but finds no more thermals than the follower.

Not supported

- that the group *causes* the effect: the strata and the same-flight windows fix pilot, day, site, hour, zone and the drift of the flight, not what brings pilots together (a line, a ridge, a route) nor the coordination needed to stay in contact;
- that larger groups straighten the progress, up to six pilots;
- that a contact means interaction (shared air within sight);
- the glide statistics are partly by definition (aligned headings in glide);
- phases exist on 1 Hz flights only; the same analyses on the xc-dynamics tracks are open.

A How H and H_{loc} are computed

$\mathrm{MSD}(\tau) \propto \tau^{2H}$, so $\log_{10}\mathrm{MSD} = 2H\log_{10}\tau + c$. H is half the slope of the least-squares line over all 21 lags (equal weight per lag). $H_{\mathrm{loc}}(\tau_c)$ is half the slope of the line fitted on the lags with $|\log_{10}\tau - \log_{10}\tau_c| \le 0.25$ (a factor 1.78 on each side), equal weight; where fewer than 3 lags lie in the window, the 3 nearest are used. It is a windowed regression, not a pointwise derivative. ΔH and ΔH_{loc} of two curves are the differences of the fitted values; for a stratified average they are the same regression on the averaged $\Delta\log\mathrm{MSD}$ (linear: slope of a difference = difference of slopes). Bands: the same computation on each bootstrap draw, percentiles lag by lag.

τ (s)	lags used	τ (s)	lags used	τ (s)	lags used
10	10–30 (3 nearest)	90	60–140	320	210–470
20	20–40 (3 nearest)	100	60–170	390	260–580
30	20–50	110	80–170	470	320–710
40	30–60	140	80–210	580	390–1000
50	30–80	170	100–260	710	470–1000
60	40–100	210	140–320	870	580–1000
80	50–140	260	170–390	1000	580–1000

Table 7: Lags of the window of each H_{loc} value. At 10 and 20 s the window is not centred; at 870 and 1000 s it is cut at 1000 s and the two values use the same lags, so the end of the curve flattens by construction. The window resolves nothing narrower than about half a decade.

Phase MSD figures (not in this report) use a different local exponent: the finite-difference derivative of $\log\mathrm{MSD}$ against $\log\Delta$ on 100 lags, without the factor $1/2$ (about 2 for ballistic motion).

Ratios and the glide ratio. Ratios are sums of numerators over sums of denominators over the runs of a population (a run goes to the stretch that holds its middle). In a small stratum a glide in rising air has almost no height lost, the glide ratio diverges and its average over strata is unstable; stratified contrasts use the *height lost per km of glide*, the inverse. Pooled tables keep the glide ratio.

B Checks of the viewer and of the analyses

The checks of the grouping itself, its code against the note and the properties of its tables, are in section 8 of the note.

- **Viewer.** The *Group flights* tab reads `groups/` of its data folder: its seven tables are **byte-identical** to those of the chosen run; the tracks it draws (`SSD_DATA`) have the same fixes, times, positions and trimmed starts as those the grouping read (`SSD_DISANTE`).
- **Analyses.** Tests on synthetic flights for every step, and on every number of the README of the analyses; the decomposition is checked on a synthetic case where the day raises a statistic by 10 and the group by 2: pooled 10, stratified 2.

C Which statistics are drawn

Unit: a population on a site-day with at least 25 flights (119 units). A statistic is redundant when the others explain most of it across units (R^2 of its linear regression on them); backward elimination drops the most explained until every $R^2 \le 0.5$. The four shares of time sum to 100%, so one of them goes first.

statistic	status	R^2 on the five drawn
time in climb (%)	drawn	0.23
searches that end in a climb (%)	drawn	0.41
climb rate (m/s)	drawn	0.20
glide speed (km/h)	drawn	0.26
glide ratio	drawn	0.38
time in transition (%)	dropped, number 1, R^2 1.00 when dropped	0.83
height gained per climb (m)	dropped, number 2, R^2 0.98 when dropped	0.91
climbs that start after a search (%)	dropped, number 3, R^2 0.93 when dropped	0.77
cross-country speed over 30 min (km/h)	dropped, number 4, R^2 0.93 when dropped	0.90
time in a search in vain (%)	dropped, number 5, R^2 0.88 when dropped	0.71
climbs per hour	dropped, number 6, R^2 0.72 when dropped	0.71
time in a search that ends in a climb (%)	dropped, number 7, R^2 0.65 when dropped	0.65

Table 8: Five statistics remain; the others are explained by them.

D Pooled curves, hang gliders and sizes

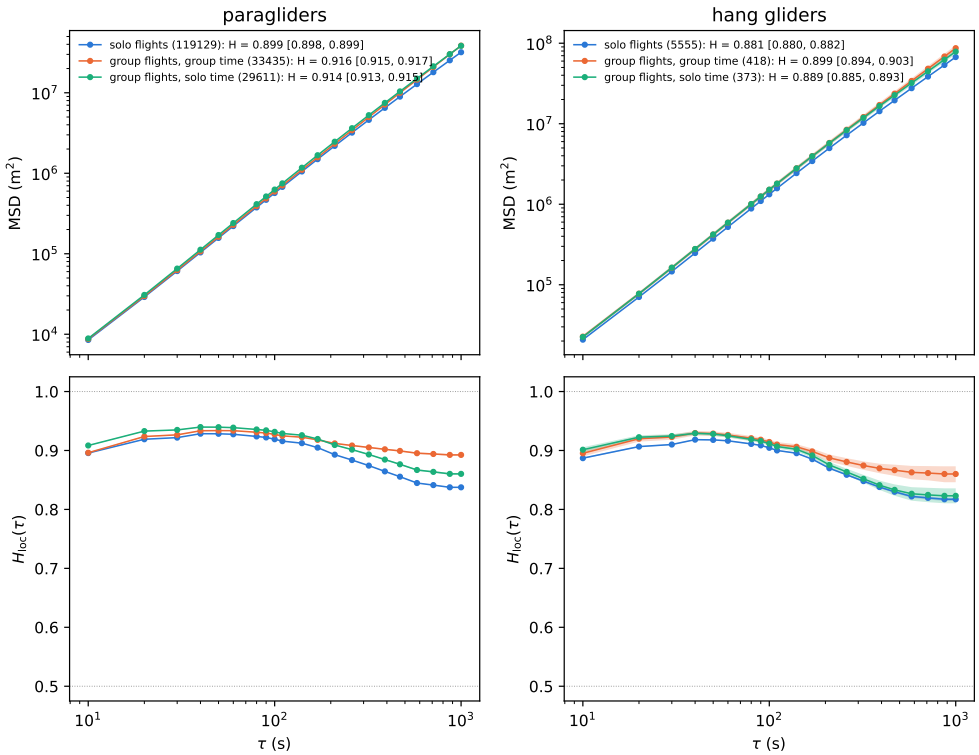


Figure 8: TA-MSD and H_{loc} of S, G and A, paragliders and hang gliders (hang gliders: 418 group flights, wide bands at long lags).

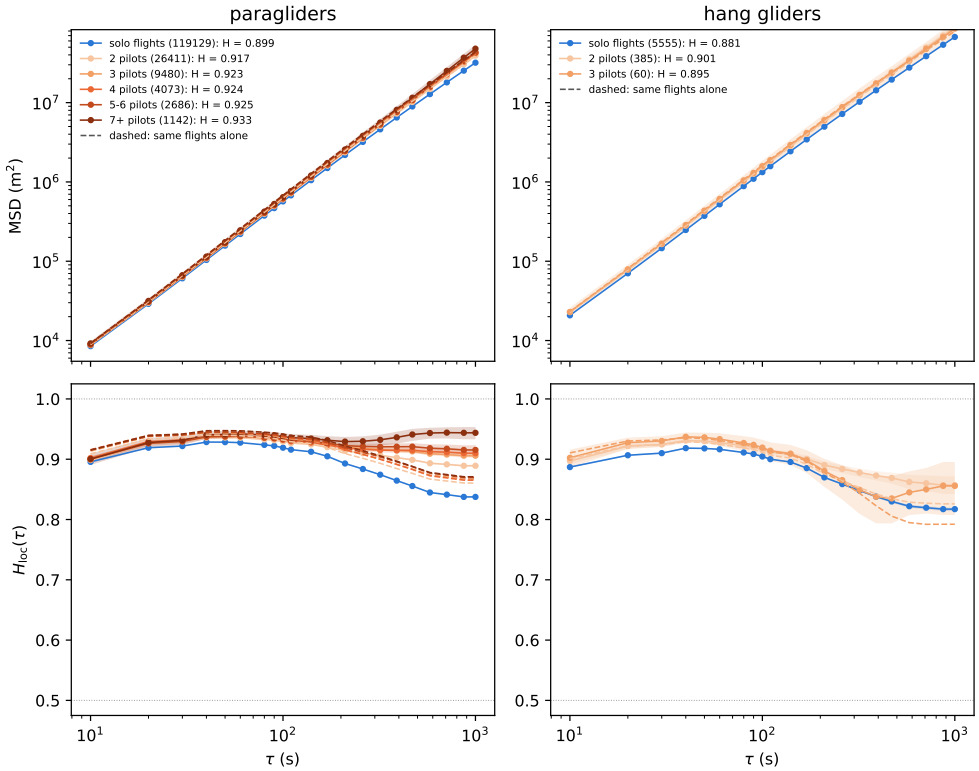


Figure 9: By the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

E Single site-days

Six site-days chosen by a fixed rule (most sizes with 15 flights or more, then most flights, at most two per takeoff site). Bands resample flights (too narrow: flights of one group are not independent).

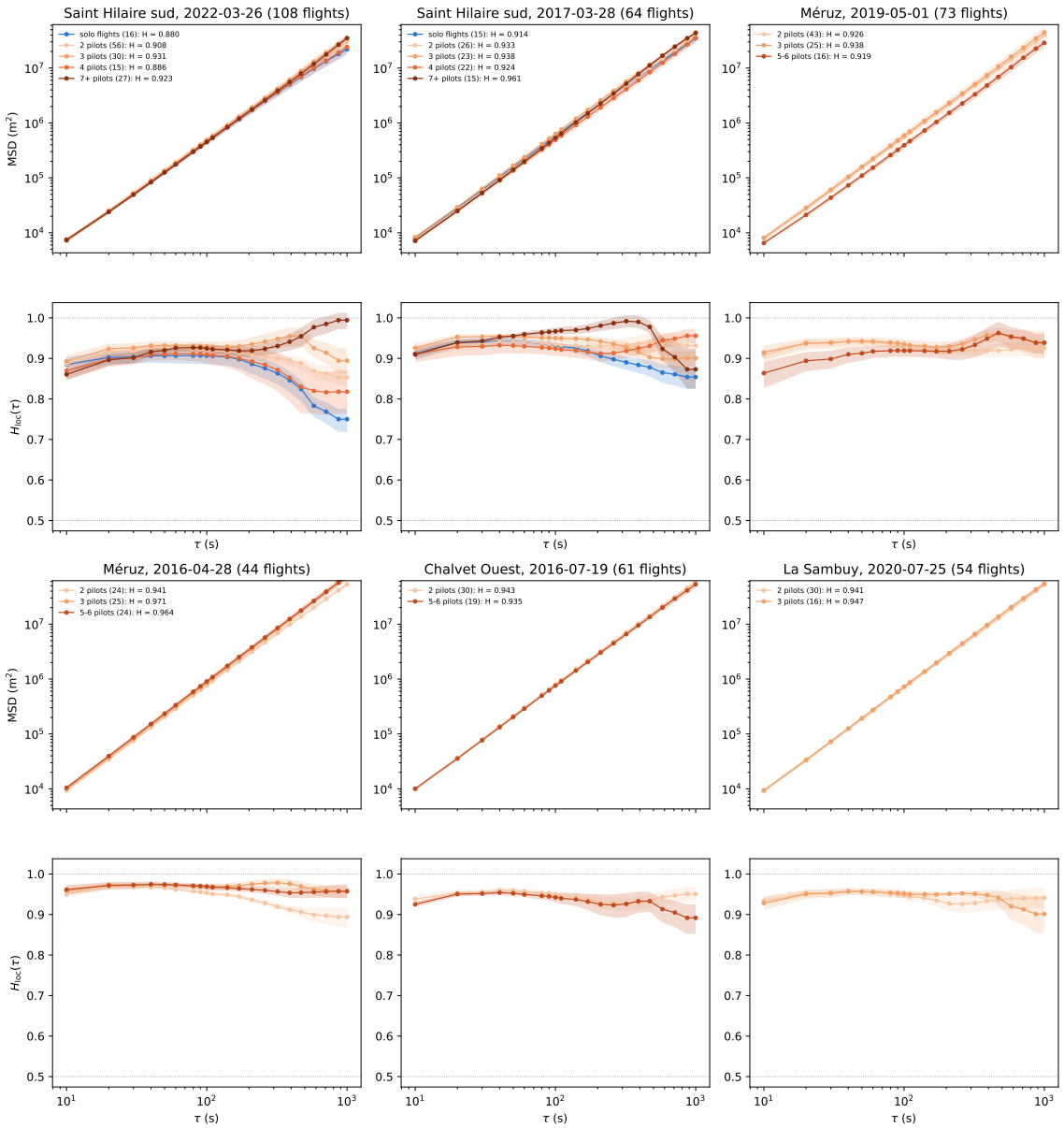


Figure 10: TA-MSD and H_{loc} on six site-days by the size of the configuration, with the solo flights of the same site-day where they reach 15 flights.

site-day	solo	2	3	4	5–6	7+
Saint Hilaire sud, 2022-03-26	0.750	0.853	0.894	0.818		0.994
Saint Hilaire sud, 2017-03-28	0.854	0.931	0.901	0.956		0.873
Méruz, 2019-05-01		0.926	0.939		0.938	
Méruz, 2016-04-28		0.894	0.958		0.957	
Chalvet Ouest, 2016-07-19		0.951			0.892	
La Sambuy, 2020-07-25		0.941	0.901			

Table 9: $H_{\text{loc}}(1000\text{ s})$ by size on each site-day (empty: fewer than 15 flights). Differences between days exceed those between sizes (the climb rate goes from 0.79 to 2.11 m/s across the six days).

F Robustness to the threshold

With $q = 0.2$ contacts need to last only 26.4 min: more and shorter groups, up to one contact in five chance.

	$q = 0.1$, $\tau^* = 38.1$ min	$q = 0.2$, $\tau^* = 26.4$ min
group flights (TA-MSD)	33 435	48 777
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, S	0.837	0.836
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, A	0.860	0.854
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, G	0.893	0.892
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same flight	+0.037 [+0.035, +0.039]	+0.032 [+0.029, +0.035]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same s-d-h	+0.043 [+0.040, +0.046]	+0.042 [+0.040, +0.044]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, A − S same s-d-h	+0.003 [+0.001, +0.006]	+0.002 [−0.001, +0.005]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same flight	−0.009 [−0.009, −0.008]	−0.016 [−0.017, −0.015]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same s-d-h	+0.002 [+0.001, +0.002]	−0.004 [−0.005, −0.003]
Δ searches to climb, G − A same s-d-h (points)	+2.6 [+2.3, +3.0]	+3.4 [+3.0, +3.7]
Δ climb rate, G − A same s-d-h (m/s)	+0.029 [+0.025, +0.034]	+0.024 [+0.020, +0.029]
Δ height lost per km, G − A same s-d-h (m/km)	−9.5 [−10.5, −8.6]	−11.5 [−12.2, −10.9]
Δ cross-country speed, G − A same s-d-h (km/h)	+1.29 [+1.16, +1.41]	+1.14 [+1.01, +1.30]

Table 10: The main results with the two thresholds (s-d-h: same site-day and hour). Every conclusion keeps its sign; at 100 s both estimates of the group effect are slightly negative with $q = 0.2$: no robust effect within one phase.

G Before, during and after with short windows

Same test as Section 3.3 with windows of 21 min, the shortest a piece of the TA-MSD allows. Groups of at least 21 min enter: more events, but at 1000 s a window holds about 26 origins, so H_{loc} over the longest lags is noisy and its drift between consecutive windows is large (controls below). The fitted H, dominated by short lags, shows no change beyond the drift; the phase statistics at entry depend on the window (time in climb +3.3 points, glide speed +0.63 km/h here, against +0.7 and +0.92 with 50 min).

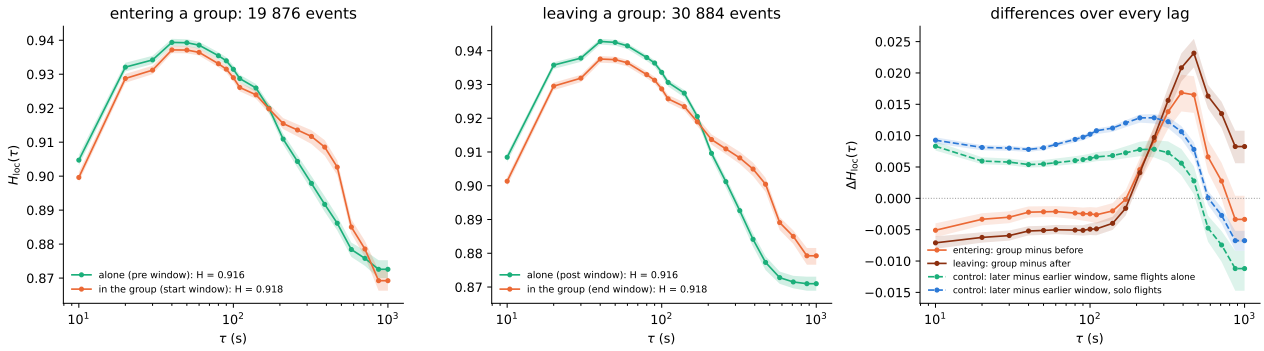


Figure 11: As Fig. 4, windows of 21 min.

comparison	events	ΔH	$\Delta H_{\mathrm{loc}}(100)$	$\Delta H_{\mathrm{loc}}(1000)$
entering: start − before	19 876	+0.002 [+0.001, +0.004]	−0.002 [−0.004, −0.001]	−0.003 [−0.007, +0.000]
leaving: end − after	30 884	+0.002 [+0.001, +0.003]	−0.005 [−0.006, −0.004]	+0.008 [+0.006, +0.011]
after − before	14 482	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.008]	+0.002 [−0.001, +0.006]
control A: later − earlier	33 658	+0.006 [+0.005, +0.006]	+0.006 [+0.006, +0.007]	−0.011 [−0.015, −0.008]
control S: later − earlier	107 324	+0.009 [+0.009, +0.010]	+0.010 [+0.010, +0.011]	−0.007 [−0.008, −0.005]

Table 11: As Table 4, windows of 21 min.

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	14 798	+3.3 [+2.9, +3.8]	+1.2 [+0.3, +2.2]	+0.043 [+0.035, +0.053]	+0.63 [+0.45, +0.77]	−8.9 [−12.1, −6.2]
leaving: end − after	23 014	+2.7 [+2.3, +3.1]	+0.9 [+0.2, +1.6]	+0.008 [+0.000, +0.018]	+0.54 [+0.42, +0.67]	−13.1 [−14.5, −11.5]
after − before	10 898	−1.5 [−2.0, −1.1]	−4.1 [−5.2, −2.9]	+0.017 [−0.001, +0.030]	+0.17 [−0.01, +0.35]	+9.6 [+5.7, +12.9]
control A: later − earlier	25 121	−2.1 [−2.4, −1.8]	−2.8 [−3.5, −2.0]	−0.001 [−0.009, +0.006]	−0.29 [−0.40, −0.15]	+3.4 [+1.9, +5.2]
control S: later − earlier	75 984	−2.0 [−2.2, −1.8]	−3.0 [−3.5, −2.7]	−0.012 [−0.017, −0.009]	−0.31 [−0.37, −0.24]	+6.2 [+5.0, +7.5]

Table 12: As Table 5, windows of 21 min.

H Leader minus follower

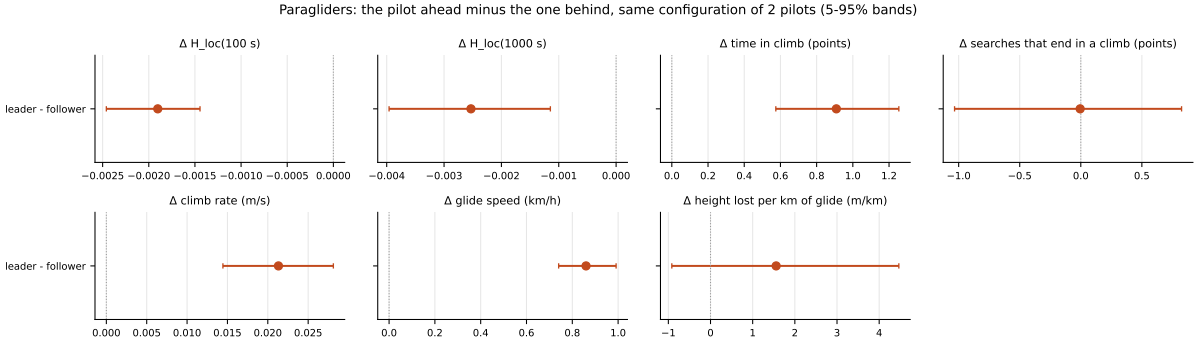


Figure 12: Leader minus follower, same configuration of two pilots.

Δ	leader − follower
$H_{\text{loc}}(100\ \text{s})$	−0.002 [−0.002, −0.001]
$H_{\text{loc}}(1000\ \text{s})$	−0.003 [−0.004, −0.001]
time in climb (%)	+0.91 [+0.57, +1.25]
searches ending in a climb (%)	−0.01 [−1.03, +0.83]
climb rate (m/s)	+0.021 [+0.014, +0.028]
glide speed (km/h)	+0.86 [+0.74, +0.99]
height lost per km of glide (m/km)	+1.6 [−0.9, +4.5]
cross-country speed (km/h)	+0.14 [+0.11, +0.16]

Table 13: Leader minus follower, same configuration of two pilots (difference with 5–95% band).

I Reproducing the report

From `xc-lab/analysis/groups/group-identification`, SSD_DISANTE mounted:

```
uv run python tamsd.py; uv run python phase_msd.py; uv run python survival.py
uv run python budget.py; uv run python performance.py; uv run python blocks.py
uv run python redundancy.py; uv run python size.py; uv run python decompose.py
uv run python zones.py; uv run python prepost.py; uv run python prepost.py --window 1260
uv run python mechanisms.py; uv run python report/facts.py
uv run python report/tables.py; cd report; latexmk -pdf group_vs_solo.tex
```

Heavy pieces stay on SSD_DISANTE, in the `analyses/` folder of each run; figures and JSON in `out/<run>/`.